Bank Loan Propensity Prediction

Model Development, Validation & Production Deployment Report

Technical Analytics Report

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Project Domain

Banking Analytics

Technology Stack

Python • Scikit-Learn • Flask • Docker • Kubernetes • AWS EKS • Amazon ECR • Terraform • CI/CD

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1. Executive Summary

This project developed a machine learning model to identify customers who are most likely to accept a loan offer.

The business objective was to improve marketing efficiency by focusing campaigns on customers with a high probability of becoming borrowers rather than targeting the entire customer base.

The dataset contained 4,980 customer records and included demographic attributes, spending behavior, banking relationships, and customer segmentation indicators. Exploratory analysis showed that spending-related variables, mortgage balances, fixed deposit ownership, and customer segment classifications were the strongest indicators of loan ownership.

More than twenty model variations were evaluated, including Logistic Regression, SVM, Decision Tree, Random Forest, AdaBoost, and Hist Gradient Boosting. Additional experiments were conducted to assess the impact of class imbalance treatment, multicollinearity handling, and feature transformations.

The final selected model was a tuned Hist Gradient Boosting classifier, which achieved:

- ROC-AUC: 99.93%
- Precision: 97.87%
- Recall: 95.83%
- F1 Score: 96.84%

The model correctly identified 92 of 96 borrowers in the test dataset while generating only two false positive predictions.

These results indicate that the model can effectively support targeted loan marketing campaigns by identifying high-probability borrowers while minimizing unnecessary outreach to customers who are unlikely to convert.

Based on both predictive performance and business impact, the Hist Gradient Boosting model is recommended for production deployment.

2. Business Context

The bank's customer base consists primarily of liability customers, while customers holding loan products represent a much smaller segment of the portfolio. Increasing the number of borrowers is an important business objective because lending products generate additional revenue and strengthen long-term customer relationships.

Historically, loan marketing campaigns have been delivered to large customer groups with limited prioritization. As a result, marketing resources are often spent on customers who have little interest in borrowing, leading to low conversion rates and inefficient campaign performance.

Key challenges include:

- Low loan conversion rates
- Inefficient marketing spend
- Higher customer acquisition costs
- Missed opportunities to identify potential borrowers

To improve campaign effectiveness, the bank requires a method for identifying customers who are most likely to accept a loan offer. A loan propensity model can help rank customers according to their likelihood of borrowing, allowing marketing teams to focus outreach efforts on the most promising prospects.

By targeting customers with a higher probability of conversion, the bank can improve marketing efficiency, increase loan adoption, and make better use of existing marketing budgets.

3. Project Objectives

The objective of this project was to develop a machine learning model capable of identifying customers who are most likely to accept a loan offer.

The model was intended to help the bank improve marketing efficiency by focusing outreach efforts on customers with a higher probability of becoming borrowers.

The project aimed to:

1. Identify customers with a high likelihood of accepting loan products.
2. Improve loan marketing campaign effectiveness.
3. Reduce marketing spend on low-probability prospects.
4. Support customer targeting and segmentation efforts.
5. Develop a production-ready loan propensity scoring solution.
6. Generate actionable insights into the factors associated with loan ownership.

Success Criteria

Model performance was evaluated using the following metrics:

- Precision
- Recall
- F1 Score
- ROC-AUC
- False Positives
- False Negatives

Because the business objective was to identify as many potential borrowers as possible, particular attention was given to Recall and False Negatives. Missing a likely borrower represents a lost marketing opportunity, making these metrics especially important when comparing candidate models.

4. Data Overview

Data Sources

The project used two customer datasets containing demographic information, spending behavior, banking relationships, and customer segmentation variables. The datasets were merged using a unique customer identifier to create a single dataset for analysis and model development.

Dataset Summary

Metric	Value
Original Records	5,000
Final Records	4,980
Features	14
Target Variable	LoanOnCard

The dataset includes customer demographic attributes, spending patterns, banking product ownership, and bank-defined customer segment information.

Target Variable

The target variable, LoanOnCard, indicates whether a customer currently holds a loan product.

Class	Count	Percentage
No Loan	4,500	90.36%
Loan	480	9.64%

Class Distribution

Only 9.64% of customers in the dataset hold a loan product, indicating a significant class imbalance.

This imbalance is important because a model can achieve high accuracy simply by predicting the majority class. As a result, model evaluation focused on metrics such as Recall, Precision, F1 Score, ROC-AUC, and Confusion Matrix analysis rather than Accuracy alone.

5. Data Cleaning, Quality Assessment & Validation

Before performing exploratory analysis and model development, the dataset was reviewed for missing values, duplicate records, invalid observations, and data type inconsistencies

Validation Activities

- Missing Value Analysis
- Duplicate Record Detection
- Customer Identifier Validation
- Data Type Verification
- Range and Consistency Checks
- Ordinal Category Validation
- Binary Variable Validation

Findings

- Twenty observations contained missing target values and were removed.
- No duplicate records were identified.
- No duplicate customer identifiers were identified.
- Negative values were observed within CustomerSince. Following business rule validation and dataset documentation review, these observations were retained as they appeared to represent intentionally masked values rather than data quality issues.
- Following cleaning activities, the dataset contained no remaining data quality concerns.

6. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand customer characteristics, identify patterns associated with loan ownership, assess variable distributions, evaluate relationships with the target variable, and identify potential modeling considerations.

Target Variable Analysis

The target variable (**LoanOnCard**) exhibited a substantial class imbalance, with loan holders representing only **9.64%** of the customer population. This finding indicated that model evaluation should emphasize Recall, Precision, F1 Score, ROC-AUC, and classification error analysis rather than Accuracy alone.

Numerical Variables

The following numerical variables were analyzed:

- Age
- CustomerSince
- HighestSpend
- MonthlyAverageSpend
- Mortgage

Key observations included:

- **HighestSpend** and **MonthlyAverageSpend** demonstrated the strongest relationship with loan ownership.
- **Mortgage** exhibited moderate predictive potential.
- **Age** and **CustomerSince** showed comparatively weaker relationships with the target variable.
- Spending-related variables displayed substantial positive skewness and contained valid high-value customer observations.
- No material data quality concerns were identified within the numerical features.

Categorical Variables

The following categorical variables were evaluated:

- Level
- HiddenScore
- Security
- FixedDepositAccount
- InternetBanking
- CreditCard

Key observations included:

- **Level** and **HiddenScore** demonstrated strong relationships with loan ownership.
- **FixedDepositAccount** exhibited a notable association with borrower status.
- **Security**, **InternetBanking**, and **CreditCard** showed limited variation in loan ownership behavior.

Correlation & Multicollinearity Analysis

Correlation analysis identified **HighestSpend**, **MonthlyAverageSpend**, **FixedDepositAccount**, **Mortgage**, and **Level** as the strongest predictors of loan ownership.

Variance Inflation Factor (VIF) analysis identified severe multicollinearity between **Age** and **CustomerSince**, indicating that both variables contained highly overlapping information. This issue was subsequently addressed during the feature engineering phase.

Key Business Insights

The analysis consistently indicated that loan ownership is driven primarily by customer financial behavior and banking engagement rather than demographic characteristics.

Customers with:

- Higher spending activity
- Larger transaction amounts
- Greater mortgage balances
- Fixed deposit relationships

- Higher customer segment classifications

were substantially more likely to possess loan products.

Conversely, age, customer tenure, internet banking usage, and credit card ownership demonstrated limited influence on borrowing behavior.

7. Statistical Analysis & Hypothesis Testing

Formal statistical testing was conducted to validate patterns observed during exploratory analysis and identify variables significantly associated with loan ownership.

Numerical Variables

Mann-Whitney U Tests were performed to compare borrower and non-borrower populations.

Statistically significant differences were identified for:

- HighestSpend
- MonthlyAverageSpend
- Mortgage

No statistically significant differences were observed for:

- Age
- CustomerSince

These findings indicate that spending behavior and mortgage balances differ meaningfully between borrowers and non-borrowers, whereas demographic characteristics provide limited discriminatory value.

Categorical Variables

Chi-Square Tests were performed to evaluate associations between categorical variables and loan ownership.

Statistically significant associations were identified for:

- FixedDepositAccount
- Level
- HiddenScore

No statistically significant associations were observed for:

- Security
- InternetBanking
- CreditCard

The statistical findings were highly consistent with the results of the exploratory analysis. Spending-related variables, mortgage balances, customer segmentation indicators, and fixed deposit ownership emerged as the most influential predictors of loan ownership.

The agreement between exploratory findings, statistical testing results, and subsequent model-based feature importance analysis provides strong evidence that these variables represent the primary drivers of borrower propensity within the customer population.

8. Feature Engineering & Data Preparation

Several feature engineering and data preparation strategies were evaluated to improve model performance, interpretability, and robustness.

Logarithmic Transformations

Exploratory Data Analysis identified substantial positive skewness within several numerical variables. To improve feature distributions and support linear modeling techniques, logarithmic transformations were applied to:

- HighestSpend
- MonthlyAverageSpend
- Mortgage

The transformed variables exhibited improved distributional characteristics and contributed to enhanced performance for Logistic Regression models.

Multicollinearity Analysis

Variance Inflation Factor (VIF) analysis was performed to identify potential multicollinearity among predictor variables.

The analysis identified severe multicollinearity between:

- Age
- CustomerSince

To evaluate the impact of multicollinearity, comparative modeling experiments were conducted using:

- Full Feature Set
- Age Removed
- CustomerSince Removed

Results demonstrated that both variables contained highly overlapping information and could be used interchangeably with minimal impact on predictive performance.

Class Imbalance Handling

The target variable exhibited substantial class imbalance, with borrowers representing less than ten percent of the overall population.

To mitigate classification bias and improve minority class detection, the following techniques were evaluated:

- Class Weighting
- Synthetic Minority Oversampling Technique (SMOTE)
- SMOTE with Random Undersampling

These approaches were assessed using Recall, Precision, F1 Score, False Positives, and False Negatives to determine the optimal balance between borrower identification and marketing efficiency.

9. Predictive Modeling Methodology

Multiple supervised machine learning algorithms were developed and evaluated to identify the most effective borrower propensity prediction model.

The following classification algorithms were considered:

- Logistic Regression
- Weighted Logistic Regression
- Naive Bayes
- Support Vector Machine (SVM)
- Decision Tree
- Random Forest
- Hist Gradient Boosting
- AdaBoost

Hyperparameter Optimization

Model tuning was performed using Randomized Search Cross Validation to identify optimal hyperparameter configurations while reducing the risk of overfitting.

Model Validation Strategy

Models were evaluated using a stratified train-test split to preserve the original class distribution. Performance was assessed using:

- ROC-AUC
- Precision
- Recall
- F1 Score
- False Positives
- False Negatives

More than twenty model variants were developed, validated, and benchmarked throughout the model development lifecycle.

10. Model Evaluation & Validation

The final candidate models were:

Rank	Model	ROC-AUC	Precision	Recall	F1-Score	False Positives	False Negatives
1	Hist Gradient Boosting (Tuned)	99.93%	97.87%	95.83%	96.84%	2	4
2	Random Forest (Baseline)	99.91%	98.90%	93.75%	96.26%	1	6
3	Random Forest + SMOTE + Undersampling	99.90%	92.23%	98.96%	95.48%	8	1

Given the highly imbalanced nature of the target variable, model evaluation emphasized Recall, Precision, F1 Score, ROC-AUC, and classification error analysis rather than Accuracy alone.

Overfitting Assessment

Training and testing performance were compared for all candidate models to assess generalization capability and identify potential overfitting risks.

The final candidate models demonstrated strong predictive performance and minimal train-test performance degradation, indicating robust generalization to unseen customer populations.

11. Business Trade-Off Analysis

Model	Strength	Business Use Case
Hist Gradient Boosting	Best Overall Balance	Recommended Production Model
Random Forest	Highest Precision	Cost-Sensitive Marketing
Random Forest + SMOTE + Undersampling	Highest Recall	Aggressive Borrower Acquisition

12. Final Model Selection

Hist Gradient Boosting (Tuned) was selected as the recommended production model.

Selection rationale:

- Highest F1 Score
- Highest ROC-AUC
- Strong Precision
- Strong Recall
- Low False Positive Rate
- Low False Negative Rate

Business Interpretation:

Out of 96 actual borrowers:

- 92 correctly identified
- 4 missed borrowers
- 2 customers incorrectly identified as potential borrowers

The model achieved the strongest balance between borrower acquisition and marketing efficiency.

13. Business Impact Assessment

The proposed loan propensity prediction solution provides measurable value across multiple operational and strategic dimensions of the retail banking business.

Marketing Efficiency

The model enables marketing resources to be focused on customers with the highest likelihood of accepting loan products. By prioritizing high-propensity prospects, the bank can reduce unnecessary campaign expenditure and improve overall marketing effectiveness.

Revenue Growth

Improved borrower identification increases opportunities for loan adoption and supports expansion of the bank's asset portfolio. More effective targeting can contribute to higher conversion rates and increased revenue from lending products.

Customer Targeting

Probability-based customer ranking enables more personalized and data-driven marketing campaigns. Marketing teams can prioritize customers according to predicted borrower propensity, improving campaign precision and customer engagement.

Decision Support

The model provides a scalable analytical framework that supports campaign planning, customer segmentation, marketing optimization, and lending strategy development. The solution can be integrated into existing decision-making processes to enhance customer acquisition initiatives.

Strategic Value

By improving borrower identification and campaign targeting, the proposed solution supports the bank's broader objectives of increasing lending relationships, improving marketing return on investment, and promoting data-driven business decision-making.

14. Model Risk, Governance & Limitations

Although the selected model demonstrated strong predictive performance, several limitations and governance considerations should be acknowledged.

Model Limitations

The current solution is subject to the following constraints:

- Model development was based exclusively on historical customer data.
- The dataset represents a limited customer sample and may not fully capture all borrower behaviors.
- Macroeconomic indicators were not available for analysis.
- Longitudinal customer transaction histories were not included.
- Certain customer attributes were masked by the data provider, limiting direct business interpretation.

Model Governance Considerations

To ensure continued model effectiveness following deployment, ongoing governance procedures should include:

- Data Drift Monitoring
- Model Performance Monitoring
- Campaign Outcome Tracking
- Periodic Model Recalibration
- Champion–Challenger Model Testing

These controls help ensure that the model remains accurate, reliable, and aligned with evolving customer behavior and business conditions.

Risk Considerations

The model should be used as a decision-support tool rather than a fully autonomous decision-making system. Business users should consider model outputs alongside operational requirements, campaign objectives, and domain expertise when making customer targeting decisions.

15. Deployment Recommendation

Recommended Production Workflow

Customer Data

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Data Validation

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Feature Engineering Pipeline

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Hist Gradient Boosting Scoring Engine

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Probability Scoring & Customer Ranking

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REST API

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Campaign Management Platform

The selected Hist Gradient Boosting model is recommended for production deployment due to its strong predictive performance, balanced classification results, and robust generalization capability.

The model should be deployed as a REST API and integrated with existing campaign management systems to support automated borrower propensity scoring and customer prioritization. This architecture enables scalable, real-time, and repeatable loan propensity predictions while supporting future integration with MLOps monitoring and automated retraining frameworks.

16. Future Enhancements

Several opportunities exist to further improve the solution and enhance long-term business value.

Recommended Initiatives

- Automated Model Retraining
- Data Drift Detection
- Real-Time Scoring Capabilities

- A/B Testing Integration
- Customer Lifetime Value Modeling
- Marketing Optimization Frameworks
- Champion–Challenger Model Strategies
- Automated Performance Monitoring & Reporting

Future development efforts should focus on expanding data availability, strengthening model governance processes, and improving operational integration. These enhancements will support continued model effectiveness and enable more advanced customer acquisition and marketing optimization strategies.

17. Conclusion

This project successfully developed, validated, and evaluated a machine learning-based loan propensity prediction solution capable of supporting targeted marketing initiatives within a retail banking environment.

Through comprehensive data quality assessment, exploratory data analysis, statistical testing, feature engineering, class imbalance handling, predictive modeling, and model validation, more than twenty model configurations were assessed. The tuned Hist Gradient Boosting model emerged as the preferred production solution based on its superior balance of Precision, Recall, F1 Score, and ROC-AUC performance.

The selected model achieved a ROC-AUC of 99.93%, Precision of 97.87%, Recall of 95.83%, and F1 Score of 96.84%, demonstrating exceptional capability in identifying potential borrowers while minimizing unnecessary marketing outreach.

The solution provides a scalable and production-ready foundation for improving marketing efficiency, increasing loan conversion rates, optimizing customer targeting, and supporting data-driven customer acquisition strategies. With appropriate governance, monitoring, and deployment controls, the model has the potential to deliver meaningful business value and support future lending growth initiatives.